

SOFTWARE CONFIGURATION

FLUIDNC

FluidNC is an easy-to-use firmware that runs on an ESP32 and controls your CNC machine through G-code. It interprets commands, drives your motors, and gives you a clean web interface so you can upload files, jog the machine, and manage settings. Once installed, FluidNC provides clear steps for you to follow, all you have to do is download it through the following link: <https://installer.fluidnc.com/fluidnc>

How to Upload and Run the File in FluidNC

- 1 - secure material to spoil board
- 2 - upload file
- 3 - home X and Y axis
- 4 - probe Z axis using touch plate
- 5 - ensure compressor is on and 70% isopropyl alcohol is loaded (if cutting aluminum)
- 6 - Start job, wait for mist, then press play



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Instructions

You'll need to follow the instructions here to make sure limit switches and homing are set up properly for your FRCNC

http://wiki.fluidnc.com/en/config/homing_and_limit_switches

The \$limits command is super useful (be sure to use this to test each switch manually before attempting to home an axis automatically)

Also: FRCNC uses two motors to drive the X axis. Make sure you have the X0 and X1 limit switches wired correctly to correspond with motor0 and motor1 for the X axis.

Also, be sure to check that your X and Y axes are square (Bryan can explain how to do this) and follow the instructions for Axis Squaring to correct any imperfections. This should be repeated any time you disassemble any part of the X or Y axis.

